

Power Supply

5V_Only!!!

LiPo Battery Charger

3.7V LiPo Battery

External Power Sense

Battery Measurement

ADC0toEXT

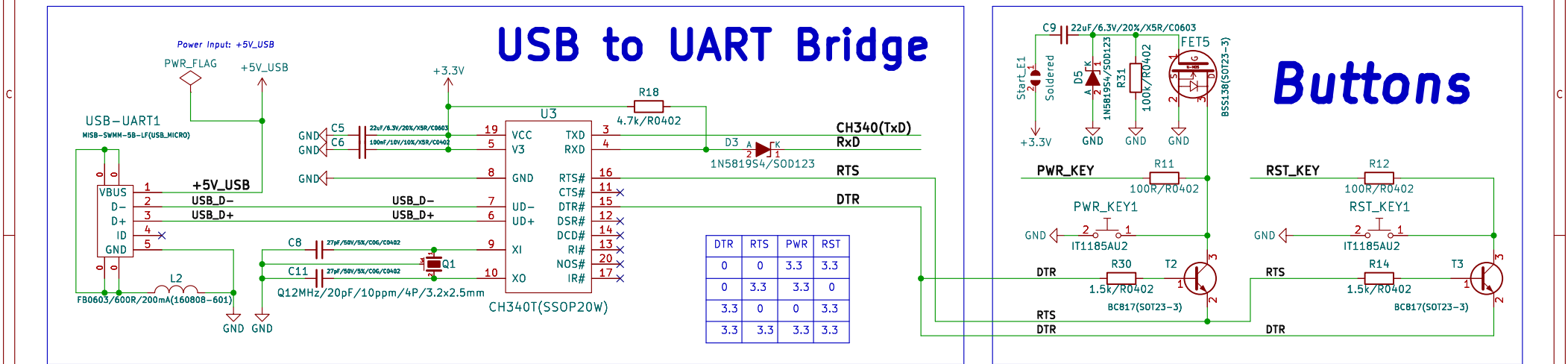
ADC0

CT5_AUX/PWR-SENS

Warning: The maximum input voltage of the 10-bit ADC0 is 1.4V!

Warning: The maximum GPIO input voltage must not exceed 1.8V!

Formula: $V_{out} = 0.6 * (1 + R_a / R_b)$



Buttons

Extensions
Power Inputs: +5V_EXT and VBAT

VBAT +5V_EXT EXT1
VCCB(3.3V) 1
GND 2
Rx D-DBG 3
Tx D-DBG 4
ADCtoEXT 5
PWR_KEY 6
RST_KEY 7
PSM_EINTn 8
HN1x10

EXT2
1 GPIO1
2 GPIO2
3 GPIO3
4 GPIO4
5 GPIO5
6 Netlight
7 RTS-AUX
8 CTS-AUX
9 Tx D-AUX
10 Rx D-AUX
HN1x10

EXT3
1 I2C0-SCL(RI)
2 I2C0-SDA(DCD)
3 SPI-MISO
4 SPI-MOSI
5 SPI-SCLK
6 SPI-CS
HN1x6

EXT4
1 Rx D
2 Tx D
HN1x3

UEXT

+3.3V
R13 1M/R0402
SPI-CS
GND
Tx D-AUX 1
I2C0-SCL(RI) 2
SPI-MISO 3
SPI-SCLK 4
mUEXT1
Rx D-AUX 5
I2C0-SDA(DCD) 6
SPI-MOSI 7
SPI-CS 8
HAB-VCR-010-LF

Voltage-Level Translators

3.3V/VCCB1 Soldered/Opened
+3.3V VCCB(3.3V)
T1 BSS138(SOT23-3)
R1 100k/R0402
FET1 WPM2015-3/TR
VCCB
R15 4.7k/R0402
+1.8V
Tx D_MAIN
CH340(TxD)
+1.8V
RxD_MAIN
RxD
Tx D
PWR_FLAG
PWR_FLAG

Warning: !!!Attention!!! Levels!!!

Extensions
Power Inputs: +5V_EXT and VBAT

VBAT +5V_EXT EXT1
VCCB(3.3V) 1
GND 2
Rx D-DBG 3
Tx D-DBG 4
ADCtoEXT 5
PWR_KEY 6
RST_KEY 7
PSM_EINTn 8
HN1x10

EXT2
1 GPIO1
2 GPIO2
3 GPIO3
4 GPIO4
5 GPIO5
6 Netlight
7 RTS-AUX
8 CTS-AUX
9 Tx D-AUX
10 Rx D-AUX
HN1x10

EXT3
1 I2C0-SCL(RI)
2 I2C0-SDA(DCD)
3 SPI-MISO
4 SPI-MOSI
5 SPI-SCLK
6 SPI-CS
HN1x6

EXT4
1 Rx D
2 Tx D
HN1x3

UEXT

+3.3V
R13 1M/R0402
SPI-CS
GND
Tx D-AUX 1
I2C0-SCL(RI) 2
SPI-MISO 3
SPI-SCLK 4
mUEXT1
Rx D-AUX 5
I2C0-SDA(DCD) 6
SPI-MOSI 7
SPI-CS 8
HAB-VCR-010-LF

Voltage-Level Translators

3.3V/VCCB1 Soldered/Opened
+3.3V VCCB(3.3V)
T1 BSS138(SOT23-3)
R1 100k/R0402
FET1 WPM2015-3/TR
VCCB
R15 4.7k/R0402
+1.8V
Tx D_MAIN
CH340(TxD)
+1.8V
Rx D_MAIN
R17 1.5k/R0402
PWR_FLAG
PWR_FLAG
+1.8V
D4 BAT54A(SOT23-3)
Tx D

Warning: !!!Attention!!! Levels!!!

Extensions
Power Inputs: +5V_EXT and VBAT

VBAT +5V_EXT EXT1
VCCB(3.3V) 1
GND 2
Rx D-DBG 3
Tx D-DBG 4
ADCtoEXT 5
PWR_KEY 6
RST_KEY 7
PSM_EINTn 8
HN1x10

EXT2
1 GPIO1
2 GPIO2
3 GPIO3
4 GPIO4
5 GPIO5
6 Netlight
7 RTS-AUX
8 CTS-AUX
9 Tx D-AUX
10 Rx D-AUX
HN1x10

EXT3
1 I2C0-SCL(RI)
2 I2C0-SDA(DCD)
3 SPI-MISO
4 SPI-MOSI
5 SPI-SCLK
6 SPI-CS
HN1x6

EXT4
1 Rx D
2 Tx D
HN1x3

UEXT

+3.3V
R13 1M/R0402
SPI-CS
GND
Tx D-AUX 1
I2C0-SCL(RI) 2
SPI-MISO 3
SPI-SCLK 4
mUEXT1
2 Rx D-AUX
4 I2C0-SDA(DCD)
6 SPI-MOSI
8 SPI-MOSI
10 SPI-CS
HAB-VCR-010-LF

Voltage-Level Translators

T1 BSS138(SOT23-3)
R1 100k/R0402
FET1 WPM2015-3/TR
VCCB
R15 4.7k/R0402
Tx D_MAIN
CH340(TxD)
+1.8V
R17 1.5k/R0402
Rx D_MAIN
PWR_FLAG
PWR_FLAG
+1.8V

Warning: !!!Attention!!! Levels!!!

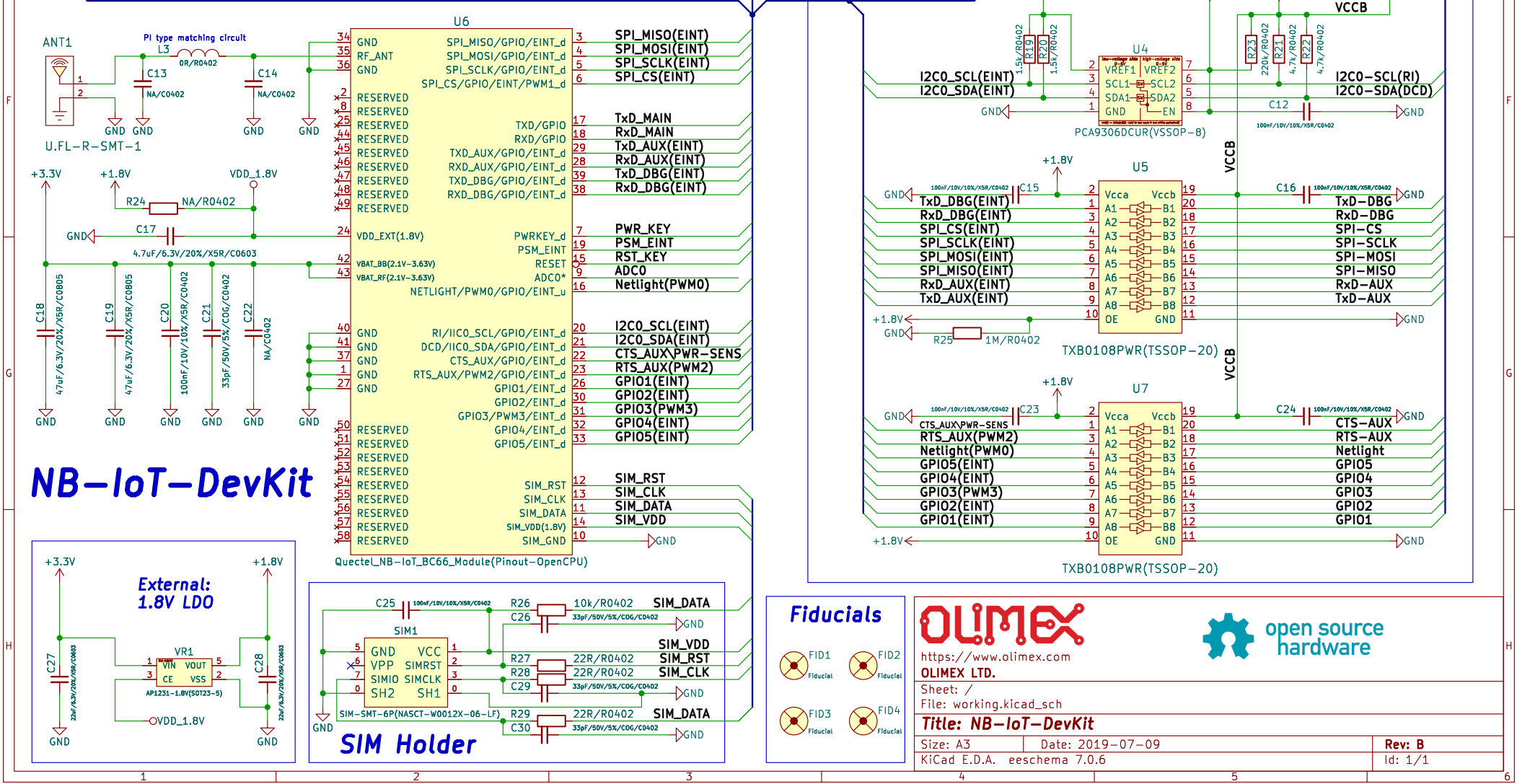
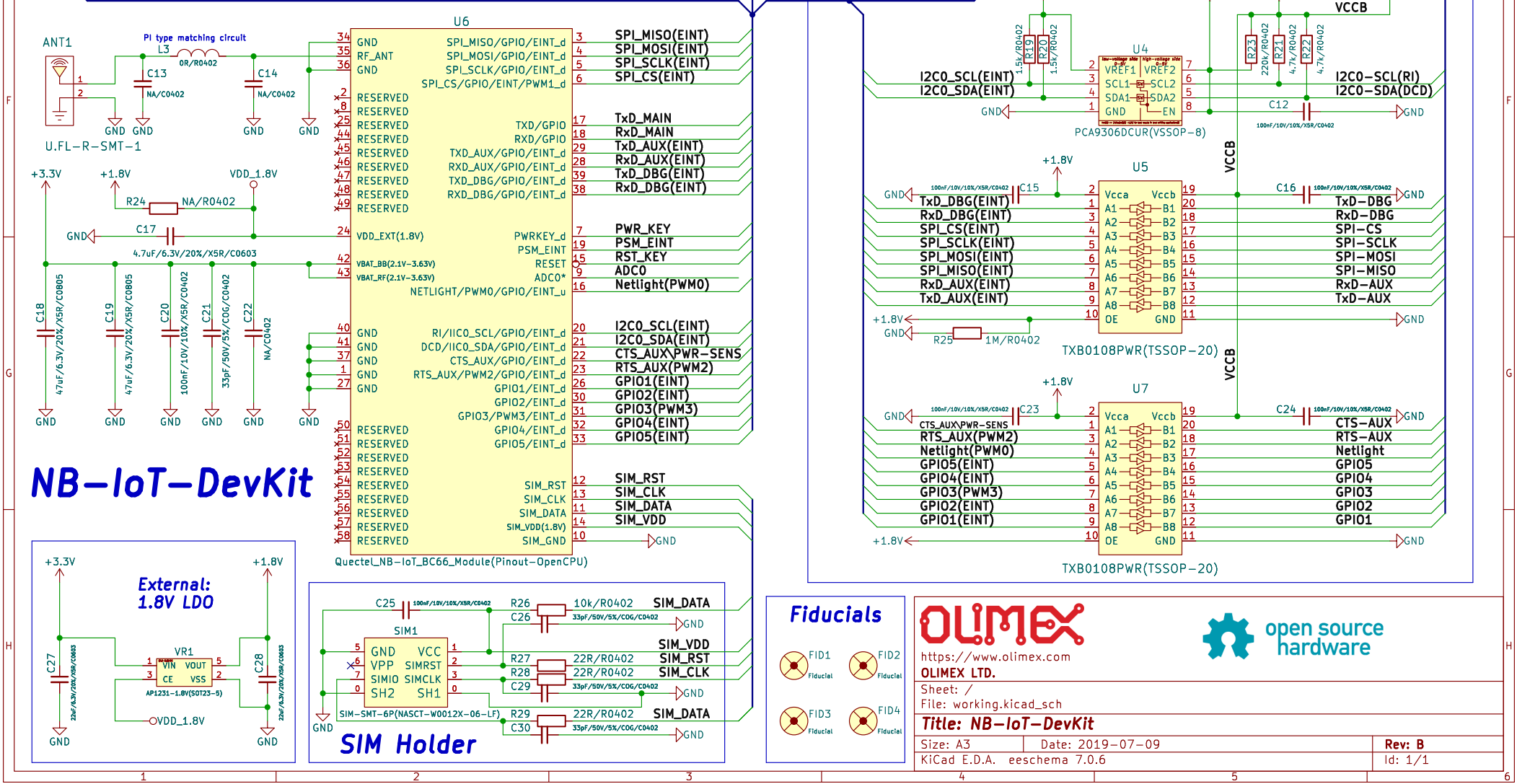
NB-IoT-DevKit

External: 1.8V LDO

SIM Holder

Fiducials

OLIMEX
open source hardware
<https://www.olimex.com>
OLIMEX LTD.
Sheet: /
File: working.kicad_sch
Title: NB-IoT-DevKit
Size: A3 Date: 2019-07-09
KiCad E.D.A. eeschema 7.0.6
Rev: B
Id: 1/1

[illegible]

NB-IoT-DevKit

External: 1.8V LDO

SIM Holder

Fiducials

FID1	FID2	FID3	FID4
(Symbol)	(Symbol)	(Symbol)	(Symbol)

Olimex Ltd.
<https://www.olimex.com>
Sheet: /
File: working.kicad_sch
Title: NB-IoT-DevKit
Size: A3 Date: 2019-07-09 Rev: B
KiCad E.D.A. eeschema 7.0.6 Id: 1/1

NB-IoT-DevKit

External: 1.8V LDO

SIM Holder

Fiducials

OLIMEX
open source hardware
https://www.olimex.com
OLIMEX LTD.
Sheet: /
File: working.kicad_sch
Title: NB-IoT-DevKit
Size: A3 Date: 2019-07-09
KiCad E.D.A. eeschema 7.0.6
Rev: B
Id: 1/1

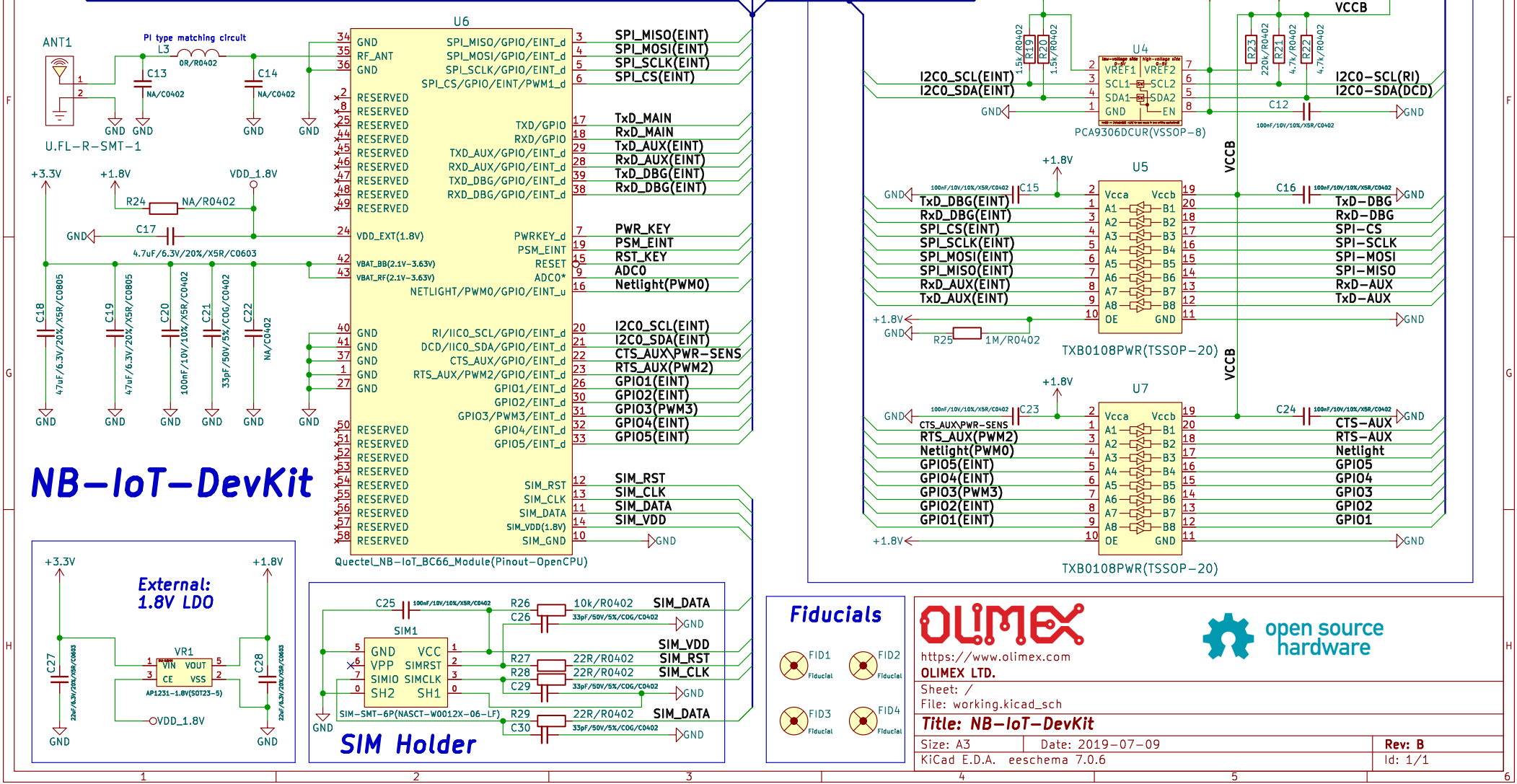
NB-IoT-DevKit

External: 1.8V LDO

SIM Holder

Fiducials

OLIMEX
open source hardware
https://www.olimex.com
OLIMEX LTD.
Sheet: /
File: working.kicad_sch
Title: NB-IoT-DevKit
Size: A3 Date: 2019-07-09
KiCad E.D.A. eeschema 7.0.6
Rev: B
Id: 1/1



ANT1

U.FL-R-SMT-1

PI type matching circuit

C13 NA/C0402

C14 NA/C0402

R24 NA/R0402

C17 4.7uF/6.3V/20%/X5R/C0603

C18 4.7uF/6.3V/20%/X5R/C0805

C19 4.7uF/6.3V/20%/X5R/C0805

C20 100nF/10V/10%/X5R/C0402

C21 100nF/10V/10%/X5R/C0402

C22 33pF/50V/5%/C0G/C0402

VDD_1.8V

U6

Quectel NB-IoT Module (Pinout - OpenCPU)

U4

PCA9306CUR(VSSOP-8)

U5

TXB0108PWR(TSSOP-20)

U7

TXB0108PWR(TSSOP-20)

SIM Holder

SIM1

SIM-SMT-6P(NASCT-W0012X-06-LF)

Fiducials

OLIMEX

open source hardware

https://www.olimex.com

OLIMEX LTD.

Sheet: /

File: working.kicad_sch

Title: NB-IoT-DevKit

Size: A3 Date: 2019-07-09

KiCad E.D.A. eeschema 7.0.6

Rev: B

Id: 1/1